
Calculators and communication devices are not allowed
Answer the following questions

1. (5 pts.) Find parametric equations of the line through the points $A(1, 2, 3)$ and $B(2, 1, 4)$.
2. (5 pts.) Show that the characteristic polynomials of a square matrix A and its transpose, A^T , are the same.
3. (5 pts.) Assuming that the stated inverses exist for a nonsingular matrix A , show that $(I + AB)^{-1}A = A(I + BA)^{-1}$.
4. (10 pts.) Show that
$$\begin{vmatrix} c_1 + 2a_1 & b_1 & a_1 \\ c_3 + 2a_3 & b_3 & a_3 \\ c_2 + 2a_2 & b_2 & a_2 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$
5. Let $W = \text{span}\{(1, 1, 0), (1, 0, 1), (2, 1, 1), (0, 1, -1)\}$.
 - (a) (7 pts.) Is $(9, 0, 9)$ in W ? Justify your answer.
 - (b) (3 pts.) Is $W = \mathbb{R}^3$? Justify your answer.
6. (5 pts.) Show that if $S = \{V_1, V_2, \dots, V_k\}$ is a linearly dependent set of vectors in a vector space V , then one of the vectors in S is a linear combination of the other vectors.
7.
 - (a) (5 pts.) Determine whether $W = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = z^2\}$ is a subspace of $\mathbb{R}^3$.
 - (b) (5 pts.) Let A, B , and C be $n \times n$ matrices. Show that if A is similar to B , and B is similar to C , then A is similar to C .
 - (c) (5 pts.) Show that similar matrices have equal determinants.
8. Let $A(1, 2, 3)$, $B(2, 1, 4)$, and $C(3, 1, 1)$ be points in $\mathbb{R}^3$.
 - (a) (10 pts.) Find an equation of the plane through A, B and C .
 - (b) (5 pts.) Find the area of the triangle ABC .
9. Let $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$.
 - (a) (5 pts.) Find the eigenvalues of A .
 - (b) (5 pts.) Find a basis for every eigenspace of A .
 - (c) (5 pts.) Find an invertible matrix P and a diagonal matrix D such that $P^{-1}AP = D$.

10. In each of the following questions, select the correct answer.

(3 pts. each)

(I) Let A be a 6×4 matrix, then the

- (A) rows of A are linearly independent
- (B) rows of A are linearly dependent
- (C) columns of A are linearly independent
- (D) columns of A are linearly dependent

(II) Let A and B be similar matrices, then

- (A) $\text{rank}(A) = \text{rank}(B)$
- (B) $\text{nullity}(A) = \text{nullity}(B)$
- (C) the eigenvalues of A and B are the same
- (D) all of the above

(III) If A and B are row equivalent matrices, then

- (A) $\det(A) = \det(B)$
- (B) the column spaces of A and B are the same
- (C) the row spaces of A and B are the same
- (D) all of the above

(IV) Let A and B be 3×3 matrices with $\det(A) = 2$ and $\det(B) = 3$. Then the $\det(2AB^{-1})$ is equal to

- (A) $4/3$
- (B) $8/3$
- (C) $16/3$
- (D) $3/16$

(V) Let A and B be $n \times n$ invertible matrices. If $\det(A) = \det(B)$, then

- (A) $A = B$.
- (B) A and B have the same eigenvalues.
- (C) A and B are similar matrices.
- (D) none of the above.

1. $U = \overrightarrow{AB} = (2 - 1, 1 - 2, 4 - 3) = (1, -1, 1) \Rightarrow$

Parametric equations are: $x = 1 + t, y = 2 - t, z = 3 + t$.

2. The Characteristic Polynomial A is $|\lambda I_n - A| = |(\lambda I_n - A)^T| = |(\lambda I_n)^T - A^T| = |\lambda I_n - A^T|$, which is the Characteristic Polynomial of A^T .

3. Solution (1): $(I + AB)^{-1}A = A(I + BA)^{-1} \iff [(I + AB)^{-1}A]^{-1} = [A(I + BA)^{-1}]^{-1} \iff A^{-1}(I + AB) = (I + BA)A^{-1} \iff A^{-1} + A^{-1}(AB) = A^{-1} + (BA)A^{-1} \iff A^{-1} + IB = A^{-1} + BI$.

Solution (2): {Actually, this is true for any matrix A , even when A is singular}.

$(I + AB)^{-1}A = A(I + BA)^{-1} \iff A = \underline{(I + AB)A(I + BA)^{-1}} \iff$

$A = \underline{(A + ABA)(I + BA)^{-1}} \iff A = \underline{A(I + BA)(I + BA)^{-1}} \iff A = AI = A$.

4. $\begin{vmatrix} c_1 + 2a_1 & b_1 & a_1 \\ c_3 + 2a_3 & b_3 & a_3 \\ c_2 + 2a_2 & b_2 & a_2 \end{vmatrix} = \begin{vmatrix} c_1 & b_1 & a_1 \\ c_3 & b_3 & a_3 \\ c_2 & b_2 & a_2 \end{vmatrix} = - \begin{vmatrix} c_1 & b_1 & a_1 \\ c_2 & b_2 & a_2 \\ c_3 & b_3 & a_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$

5. (a) $c_1(1, 1, 0) + c_2(1, 0, 1) + c_3(2, 1, 1) + c_4(0, 1, -1) = (9, 0, 9) \Rightarrow \left[\begin{array}{cccc|c} 1 & 1 & 2 & 0 & 9 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & -1 & 9 \end{array} \right] \equiv$

$\dots \equiv \left[\begin{array}{cccc|c} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & -1 & 9 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \Rightarrow c_1 = -r - s, c_2 = -r + s + 9, c_3 = r, c_4 = s.$

Yes, $(9, 0, 9) \in W$. $\{(-r - s)(1, 1, 0) + (-r + s + 9)(1, 0, 1) + r(2, 1, 1) + s(0, 1, -1) = (9, 0, 9)\}$.

(b) No, $W \neq \mathbb{R}^3$. A basis for W is $\{_1(1, 1, 0), (1, 0, 1)\} \Rightarrow \dim(W) = 2$.

6. $S = \{V_1, V_2, \dots, V_k\}$ is linearly dependent $\Rightarrow \exists c_1, c_2, \dots, c_n \in \mathbb{R}$ not all are zeros such that $0 = c_1V_1 + c_2V_2 + \dots + c_kV_k$. Let $c_j \neq 0$, then $-c_jV_j = c_1V_1 + c_2V_2 + \dots + c_{j-1}V_{j-1} + c_{j+1}V_{j+1} + \dots + c_kV_k \Rightarrow V_j = \left(\frac{c_1}{-c_j}\right)V_1 + \left(\frac{c_2}{-c_j}\right)V_2 + \dots + \left(\frac{c_{j-1}}{-c_j}\right)V_{j-1} + \left(\frac{c_{j+1}}{-c_j}\right)V_{j+1} + \dots + \left(\frac{c_k}{-c_j}\right)V_k$.

7. (a) $W = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = z^2\}$ is not a subspace of $\mathbb{R}^3$. Since, $(0, 2, 2), (1, 0, 1) \in W$, but $(0, 2, 2) + (1, 0, 1) = (1, 2, 3) \notin W$.

(b) A is similar to B , and B is similar to $C \Rightarrow \exists$ non singular matrices X and Y such that $A = X^{-1}BX$ and $B = Y^{-1}CY \Rightarrow A = X^{-1}(Y^{-1}CY)X = (X^{-1}Y^{-1})C(YX) = Z^{-1}CZ$, where $Z = YX$.

(c) Let A and B be similar matrices $\Rightarrow \exists$ non singular matrix X such that $A = X^{-1}BX \Rightarrow |A| = |X^{-1}BX| = |X^{-1}||B||X| = \frac{1}{|X|}|B||X| = |B|$.

8. (5 pts.) $\overrightarrow{AB} = (1, -1, 1), \overrightarrow{BC} = (1, 0, -3)$. Let $N = \overrightarrow{AB} \times \overrightarrow{BC} = \begin{vmatrix} i & j & k \\ 1 & -1 & 1 \\ 1 & 0 & -3 \end{vmatrix} = (3, 4, 1)$.

(a) (5 pts.) Equation of the plane: $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0 \Rightarrow 3(x - 1) + 4(y - 2) + (z - 3) = 0 \Rightarrow \boxed{3x + 4y + z - 14 = 0}$.

9. (a) $|\lambda I_3 - A| = \begin{vmatrix} \lambda - 2 & 0 & 0 \\ 0 & \lambda - 1 & -1 \\ 0 & -1 & \lambda - 1 \end{vmatrix} = (\lambda - 2) [(\lambda - 1)^2 - 1] = (\lambda - 2) (\lambda^2 - 2\lambda) = \lambda (\lambda - 2)^2$. Let $\lambda_1 = \lambda_2 = 2, \lambda_3 = 0$.

(b) Solve: $((2) I_3 - A) X = O$, $\left[\begin{array}{ccc|c} 2-2 & 0 & 0 & 0 \\ 0 & 2-1 & -1 & 0 \\ 0 & -1 & 2-1 & 0 \end{array} \right] = \left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \end{array} \right] \equiv$
 $\dots \equiv \left[\begin{array}{ccc|c} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \Rightarrow$

The eigenvectors corresponding to the eigenvalue 2, are $X = r \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \neq O$,
i.e., $(r^2 + s^2 \neq 0)$

Solve: $((0) I_3 - A) X = O$, $\left[\begin{array}{ccc|c} -2 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & -1 & -1 & 0 \end{array} \right] \equiv \dots \equiv \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \Rightarrow$

The eigenvectors corresponding to the eigenvalue 0, are $X = r \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \neq O$, i.e.,
 $r \neq 0$.

(c) Take $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, and $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix}$. Then $P^{-1}AP = D$.

10. (a) Let A be a 6×4 matrix, then the

rows of A are linearly dependent

(b) Let A and B be similar matrices, then

all of the above

(c) If A and B are row equivalent matrices, then

the row spaces of A and B are the same

(d) Let A and B be 3×3 matrices with $\det(A) = 2$ and $\det(B) = 3$.

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(e) Let A and B be $n \times n$ invertible matrices. If $\det(A) = \det(B)$, then

none of the above